Heston Model Calibration Using Numerical Optimisation

A Practical Implementation and Analysis

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Abstract

This research note presents a complete and internally consistent calibration framework for the Heston stochastic volatility model, with particular emphasis on numerical stability, reproducibility, and practical implementation. Starting from the risk-neutral dynamics, we derive the affine characteristic function and employ Fourier-based methods for pricing European options. Model parameters are obtained by solving a constrained nonlinear least-squares problem using the Levenberg–Marquardt algorithm.

Numerical experiments conducted in a controlled synthetic setting demonstrate that the calibrated model reproduces implied volatility smiles and term structures accurately across strikes and maturities. The results highlight an important feature of Heston calibration: excellent surface-level fits may be achieved despite non-uniqueness of the calibrated parameter vector. The proposed framework provides a transparent end-to-end reference for quantitative research, model validation, and comparative studies of stochastic volatility models.

1 Introduction

Stochastic volatility models play a central role in modern option pricing and risk management, as they are capable of reproducing the volatility smiles and skews observed in liquid option markets. Among these models, the Heston stochastic volatility model [2] remains one of the most widely used in practice. Its enduring popularity stems from a favourable balance between expressive power and analytical tractability: the variance process follows a mean-reverting square-root diffusion, while the joint dynamics admit a closed-form affine characteristic function. This structure enables efficient Fourier-based pricing of European options and makes the model well suited to large-scale calibration.

Despite its theoretical elegance, practical calibration of the Heston model is a nontrivial numerical problem. The mapping from model parameters to option prices is highly nonlinear, constrained, and often ill-conditioned, particularly when fitting across multiple maturities and strikes. Objective functions may exhibit flat regions, multiple local minima, and strong parameter interdependencies, especially between the correlation, volatility-of-volatility, and mean-reversion parameters. As a consequence, naive optimisation approaches can lead to unstable or economically implausible parameter estimates, even when pricing formulas themselves are implemented correctly.

This research note addresses the calibration of the Heston model from the perspective of numerical optimisation rather than model specification alone. Starting from the risk-neutral dynamics, we derive the affine characteristic function and employ it to price European options using Fourier-based methods. We then construct a constrained nonlinear least-squares calibration framework, placing particular emphasis on parameterisation, numerical stability, and solver choice. The calibration setup is compatible with both option price data and implied volatility surfaces, and is designed to support repeated use in research, model validation, and trading-related analysis.

The contribution of this note is not the introduction of new pricing formulas or theoretical extensions of the Heston model, but the presentation of a complete, self-contained, and implementation-oriented calibration workflow. By making explicit the modelling assumptions, numerical design choices, and diagnostic considerations involved, the paper aims to bridge the gap between textbook derivations and production-level calibration routines.

The remainder of the paper is organised as follows. Section 2 introduces the Heston model under the risk-neutral measure. Section 3 derives the characteristic function, with technical details deferred to Appendix A. Section 4 describes Fourier-based pricing of European options. Section 5 presents the calibration framework and numerical optimisation strategy. Section 6 reports numerical results and diagnostic analyses. Section 7 concludes with a discussion of practical considerations and limitations.

2 The Heston Model

Under the risk-neutral measure $\mathbb{Q}$, the Heston stochastic volatility model specifies the joint dynamics of the asset price S_t and its instantaneous variance v_t as

$$dS_t = (r - q)\, S_t\, dt + \sqrt{v_t}\, S_t\, dW_t^{(1)}, \tag{1}$$

$$dv_t = \kappa(\bar{v} - v_t)\, dt + \sigma\sqrt{v_t}\, dW_t^{(2)}, \tag{2}$$

where r denotes the constant risk-free rate, q the continuous dividend yield, and $W_t^{(1)}$ and $W_t^{(2)}$ are Brownian motions with instantaneous correlation

$$dW_t^{(1)}\, dW_t^{(2)} = \rho\, dt.$$

The variance process v_t follows a Cox–Ingersoll–Ross (CIR) square-root diffusion, ensuring non-negativity under standard conditions. The parameters have the following interpretations: $\kappa > 0$ controls the speed of mean reversion of variance toward the long-run level $\bar{v} > 0$, $\sigma > 0$ determines the volatility of variance (volatility-of-volatility), $\rho \in (-1, 1)$ captures the correlation between asset returns and variance innovations, and $v_0 > 0$ denotes the initial variance.

The correlation parameter ρ plays a central role in generating implied volatility skew: negative values, commonly observed in equity markets, induce a leverage effect whereby downward price moves are associated with increases in variance. The parameters $(\kappa, \bar{v}, v_0)$ jointly determine the term structure of variance and hence the maturity dependence of implied volatility.

A sufficient condition for strict positivity of the variance process is given by the Feller condition,

$$2\kappa\bar{v} > \sigma^2, \tag{3}$$

although in practical calibration this condition is sometimes relaxed. Whether the Feller condition is enforced or treated as a soft constraint is a modelling choice with numerical implications, discussed later in the calibration section.

The affine structure of the Heston model implies that the conditional characteristic function of $\log S_T$ is available in closed form. This property underpins efficient Fourier-based pricing of European options and forms the basis for the calibration procedures developed in subsequent sections.

3 Characteristic Function

A key advantage of the Heston model is that the logarithm of the asset price admits an affine characteristic function under the risk-neutral measure. Let

$$X_T := \log S_T.$$

The conditional characteristic function of X_T given (S_0, v_0) is defined as

$$\phi(u;T) := \mathbb{E}^{\mathbb{Q}}\left[e^{iuX_T}\right], \qquad u \in \mathbb{C}.$$

For the Heston model, $\phi(u;T)$ takes the exponential-affine form

$$\phi(u;T) = \exp(iu \log S_0 + A(u,T) + B(u,T)\, v_0), \tag{4}$$

where the functions $A(u,T)$ and $B(u,T)$ solve a system of Riccati-type ordinary differential equations determined by the model parameters $(\kappa, \bar{v}, \sigma, \rho)$.

Closed-form expressions for $A(u,T)$ and $B(u,T)$ are available and are well documented in the literature [2]. For completeness, we summarise the final expressions in Appendix A, where the derivation is carried out in detail. In implementation, particular care must be taken with branch choices of complex logarithms to ensure numerical stability and continuity across maturities.

The affine representation (4) enables efficient evaluation of European option prices via Fourier-based methods. In the next section, we describe how the characteristic function is used within Fourier inversion schemes to compute option prices across a grid of strikes, forming the computational core of the calibration procedure.

4 Fourier-Based Pricing of European Options

The availability of a closed-form characteristic function allows European option prices under the Heston model to be computed efficiently using Fourier inversion techniques. These methods are particularly well suited to calibration, as they permit fast and stable evaluation of option prices across a grid of strikes for a fixed maturity.

In Fourier-based approaches, European option prices may be expressed as integrals involving the characteristic function of the log-price. In general, one may write

$$C(K,T) = \int_{\mathbb{R}} \mathcal{K}(u;K,T)\, \phi(u;T)\, du, \tag{5}$$

where $\phi(u;T)$ is the characteristic function defined in Section 3 and $\mathcal{K}(u;K,T)$ denotes a formulation-dependent kernel whose explicit form depends on the chosen Fourier inversion scheme (e.g. Carr–Madan, Lewis, or COS). Throughout this paper, the specific form of $\mathcal{K}$ is determined by the pricing method under consideration and is given explicitly in the corresponding appendix.

Several equivalent Fourier pricing schemes exist in the literature. The Carr–Madan approach [4] introduces exponential damping to ensure square integrability and applies the fast Fourier transform (FFT) to recover option prices on a strike grid. Alternative formulations, such as the Lewis transform [3], lead to closely related integral representations with different numerical properties. In Appendix C, we present the Carr–Madan FFT method in detail, while Appendix B describes the COS method, which is often favoured for its accuracy and numerical stability.

Within the calibration procedure, Fourier pricing is implemented maturity-by-maturity: for each maturity T, the characteristic function is evaluated once on a fixed frequency grid, and option prices for all strikes at that maturity are obtained simultaneously. This structure significantly reduces computational cost compared to strike-by-strike pricing and makes repeated objective function evaluations feasible within nonlinear optimisation routines.

Careful selection of numerical parameters—such as integration bounds, damping factors, and grid resolution—is essential for stable calibration. These parameters are held fixed across optimisation iterations to avoid introducing spurious noise into the objective function. When consistent numerical settings are used, Fourier-based pricing provides smooth and reliable option values suitable for gradient-based and derivative-free optimisation methods alike.

All Fourier-based pricing methods require careful handling of complex logarithms and square roots appearing in the Heston characteristic function. In particular, the discriminant $d(u)$ and logarithmic terms must be evaluated using a consistent branch choice across frequencies and maturities. Inconsistent branch selection can introduce discontinuities that lead to numerical instability or spurious oscillations in FFT-based pricing. In practice, this issue is mitigated by enforcing $\mathrm{Re}(d(u)) > 0$ and by using numerically stable formulations such as the Little Trap representation.

5 Calibration Framework

The calibration of the Heston model is formulated as a constrained nonlinear optimisation problem, in which model parameters are chosen to minimise the discrepancy between observed market option data and model-implied values. Although the pricing formulas are available in semi-closed form, the calibration problem itself is numerically challenging due to parameter constraints, strong nonlinearities, and interactions among parameters.

5.1 Calibration targets and objective function

Let $\{(K_i, T_i)\}_{i=1}^N$ denote a set of strikes and maturities observed in the market on a given valuation date, and let C_i^{mkt} denote the corresponding observed European option prices (typically mid quotes). Given model prices $C^{\mathrm{model}}(K_i, T_i; \theta)$ computed via the Fourier-based pricing methods described in Section 4, calibration is formulated as a weighted nonlinear least-squares problem:

$$\min_{\theta \in \Theta} \mathcal{L}(\theta) := \sum_{i=1}^{N} w_i \left(C_i^{\mathrm{mkt}} - C^{\mathrm{model}}(K_i, T_i; \theta) \right)^2, \tag{6}$$

where $\theta = (\kappa, \bar{v}, \sigma, \rho, v_0)$ denotes the Heston parameter vector and $w_i \geq 0$ are weighting coefficients.

In practice, vega-weighted objectives are commonly employed to balance the influence of options across strikes and maturities. In this case, the weights are defined as

$$w_i = \frac{1}{\mathrm{Vega}_{\mathrm{BS}}(K_i, T_i)^2}, \tag{7}$$

where $\mathrm{Vega}_{\mathrm{BS}}(K_i, T_i)$ denotes the Black–Scholes vega evaluated at the market-implied volatility for strike K_i and maturity T_i. This weighting scheme prevents deep out-of-the-money options, which typically have small vegas, from dominating the calibration error.

Calibration may alternatively be performed in implied-volatility space by replacing prices in (6) with Black–Scholes implied volatilities. While implied-volatility objectives align naturally with market quoting conventions, price-based objectives are often numerically smoother, as they avoid a nested implied-volatility inversion at each optimisation iteration. Alternative weighting schemes, such as relative price errors or uniform weights, may also be employed depending on data quality and calibration objectives.

5.2 Parameter constraints and reparameterization

The Heston model parameters are subject to natural constraints:

$$\kappa > 0, \qquad \bar{v} > 0, \qquad \sigma > 0, \qquad v_0 > 0, \qquad \rho \in (-1, 1).$$

To handle these constraints robustly within unconstrained optimisation algorithms, we introduce a smooth reparameterization in terms of an unconstrained vector $x \in \mathbb{R}^5$:

$$\kappa = e^{x_1}, \qquad \bar{v} = e^{x_2}, \qquad \sigma = e^{x_3}, \qquad \rho = \tanh(x_4), \qquad v_0 = e^{x_5}.$$

This transformation guarantees admissible parameters for all values of x and avoids the need for hard constraints during optimisation.

In some applications, it is desirable to enforce the Feller condition

$$2\kappa\bar{v} > \sigma^2,$$

which ensures strict positivity of the variance process. Rather than imposing this condition as an inequality constraint, it may be enforced directly through reparameterization. In this case, $\bar{v}$ is defined as

$$\bar{v} = \frac{\sigma^2}{2\kappa} + e^{x_2} = \frac{(e^{x_3})^2}{2e^{x_1}} + e^{x_2}, \tag{8}$$

while κ, σ, ρ, and v_0 are parameterised as above. When the Feller condition $2\kappa\bar{v} > \sigma^2$ is violated, the variance process may reach zero with positive probability. This does not invalidate the Heston characteristic function or Fourier-based pricing methods, which remain well defined. However, violation of the Feller condition can increase numerical sensitivity and complicate simulation-based schemes. In practice, market calibration often yields parameter estimates that violate the Feller condition, and enforcing it strictly may degrade the quality of fit. For this reason, both constrained and unconstrained formulations are commonly used in practice.

This construction guarantees $2\kappa\bar{v} - \sigma^2 > 0$ for all $x_2 \in \mathbb{R}$ and avoids circular dependence by expressing $\bar{v}$ explicitly in terms of the underlying unconstrained variables. In practice, enforcing the Feller condition in this manner can improve numerical stability, although it is not strictly required for successful market calibration.

5.3 Optimisation algorithm

The calibration problem (6) is solved using the Levenberg–Marquardt algorithm, which interpolates between the Gauss–Newton method and gradient descent. This makes it well suited to nonlinear least-squares problems where the residual structure is smooth but the Jacobian may be ill-conditioned.

At each iteration, the algorithm updates the parameter vector by solving a damped linear system involving the Jacobian of pricing residuals with respect to the transformed parameters. Gradients may be computed via finite differences or complex-step differentiation, provided that the pricing routine is implemented with sufficient numerical smoothness.

Because the objective function may exhibit multiple local minima, the optimisation is initialised using multiple starting points drawn from economically reasonable parameter ranges. The solution with the lowest final objective value is retained. This multi-start strategy significantly improves robustness relative to single-start local optimisation. Gradients of the calibration objective may be computed using complex-step differentiation, which provides machine-precision accuracy without the subtractive cancellation errors associated with finite differences. For a smooth real-valued function $f(\theta)$, the derivative with respect to θ_j is approximated as

$$\frac{\partial f}{\partial \theta_j} \approx \frac{\operatorname{Im}[f(\theta + ihe_j)]}{h},$$

where e_j denotes the j-th unit vector and h is a small step size (typically $h \approx 10^{-20}$). This approach is particularly effective for Fourier-based pricing functions, which are analytic in the model parameters.

5.4 Numerical considerations

To ensure stable convergence, several practical considerations are observed. First, numerical parameters of the Fourier pricing routine (integration bounds, grid resolution, damping factors) are fixed throughout the optimisation to avoid introducing noise into the objective function. Second, option prices are scaled by the forward or spot level to improve conditioning. Finally, weak regularisation toward plausible parameter values may be employed when calibrating to sparse or noisy datasets, although such regularisation is kept sufficiently small so as not to dominate the market fit.

Taken together, these design choices yield a calibration framework that is numerically stable, computationally efficient, and suitable for repeated use in research and production environments.

6 Implementation Notes and Reproducibility

While the mathematical framework developed above is fully specified, practical implementation of the Heston calibration requires a number of numerical conventions and safeguards to ensure stability and reproducibility. The numerical results reported in the following section are obtained using the implementation choices outlined below.

- **Analytic continuation and branch selection.** All Fourier-based pricing methods require evaluation of the Heston characteristic function at complex arguments. In particular, consistent branch selection for complex square roots and logarithms is essential. Throughout, the discriminant $d(u)$ is evaluated using the branch satisfying $\mathrm{Re}(d(u)) > 0$, in line with standard practice.

- **Numerically stable characteristic function.** To avoid discontinuities arising from the logarithmic term in the affine solution, numerically stable representations equivalent to the *Little Trap* formulation are employed. This ensures smooth behaviour of the characteristic function across the full frequency domain required for pricing and calibration.

- **Fourier pricing parameters.** Fourier inversion integrals are evaluated over truncated frequency domains using finite grids. Integration bounds, grid sizes, and damping parameters are chosen following established practitioner guidelines and are held fixed across experiments to ensure reproducibility. While these parameters may require tuning for specific market datasets, no ad hoc adjustments are made within the reported experiments.

- **Calibration objective and optimisation.** Calibration is performed using a nonlinear least-squares objective defined either in price space or implied-volatility space. Optimisation is carried out using a Levenberg–Marquardt algorithm with numerically approximated Jacobians. This choice reflects common practice in production calibration pipelines and provides a robust trade-off between stability and computational efficiency.

- **Controlled experimental setting.** To isolate numerical behaviour from data quality effects, the numerical experiments in Section 7 are conducted using synthetic option data generated from known Heston parameters. This allows calibration accuracy, parameter non-uniqueness, and residual structure to be assessed in a controlled and fully reproducible setting.

All numerical experiments are implemented in Python using standard scientific computing libraries. The full source code required to reproduce the results, including pricing, calibration, and diagnostic plots, is provided alongside this note.

7 Numerical Experiments

This section reports numerical experiments designed to validate the calibration framework in a controlled and fully reproducible setting. Rather than using market data, which may confound numerical effects due to noise and liquidity constraints, we perform calibration on synthetic option prices generated from known Heston parameters. This allows the accuracy of the calibration procedure and the behaviour of residuals to be assessed independently of data quality issues.

European call prices are generated using the Fourier-based pricing method described in Section 4, and Black–Scholes implied volatilities are recovered via a bisection algorithm. Calibration is performed using the nonlinear least-squares objective introduced in Section 5 with fixed numerical tolerances across strikes and maturities.

7.1 Synthetic data and calibration setup

Synthetic implied volatility surfaces are generated from a reference Heston parameter vector

$$(v_0, \kappa, \bar{v}, \sigma, \rho) = (0.04,\ 1.5,\ 0.04,\ 0.5,\ -0.7),$$

across a grid of strikes and maturities spanning short-, medium-, and long-dated options. These synthetic implied volatilities serve as "market" observations for the calibration procedure. The parameter ordering is $(v_0, \kappa, \bar{v}, \sigma, \rho)$ throughout; for readability, table entries are reported in the conventional $(\kappa, \bar{v}, \sigma, \rho, v_0)$ order.

Model prices are computed using the Lewis (2001) Fourier inversion formula, evaluated via numerical quadrature over a truncated frequency domain. All numerical parameters (integration bounds, tolerances, and grid resolution) are held fixed throughout the experiments to ensure reproducibility.

	κ	$\bar{v}$	σ	ρ	v_0
True parameters	1.50	0.04	0.50	−0.70	0.04
Calibrated parameters	1.50	0.04	0.50	−0.70	0.04

Table 1: True and calibrated Heston parameters for the synthetic experiment.

In this noise-free synthetic experiment, the calibrated parameters recover the data-generating values up to numerical precision. This confirms the internal consistency of the pricing, implied-volatility inversion, and optimisation components of the calibration framework.

Initialisation	RMSE_σ	Converged
Start 1	$< 10^{-7}$	Yes
Start 2	$< 10^{-7}$	Yes
Start 3	$< 10^{-7}$	Yes
Start 4	$< 10^{-7}$	Yes
Start 5	$< 10^{-7}$	Yes

Table 2: Multi-start calibration results for the synthetic implied volatility surface.

Metric	RMSE_σ	MAE_σ	Max Error
Value	1.5×10^{-8}	4.1×10^{-9}	7.5×10^{-8}

Table 3: Implied volatility fit metrics for the synthetic calibration experiment.

Tables 1–3 demonstrate that the calibration recovers the synthetic implied volatility surface with errors at numerical precision. The multi-start experiment (Table 2) confirms that convergence is robust with respect to initialisation in this controlled setting.

In practical calibration to market data, exact parameter recovery is neither expected nor desirable due to observation noise, bid–ask effects, and model misspecification; these effects are examined in the context of real-market calibration in future work.

7.2 Smile fit

Figure 1 compares the implied volatility smile at a representative maturity for the synthetic data and the calibrated Heston model. The calibrated smile is visually indistinguishable from the synthetic smile across the full strike range, consistent with the low implied-volatility RMSE reported in Table 3. This indicates that the dominant skew and curvature features are captured accurately.

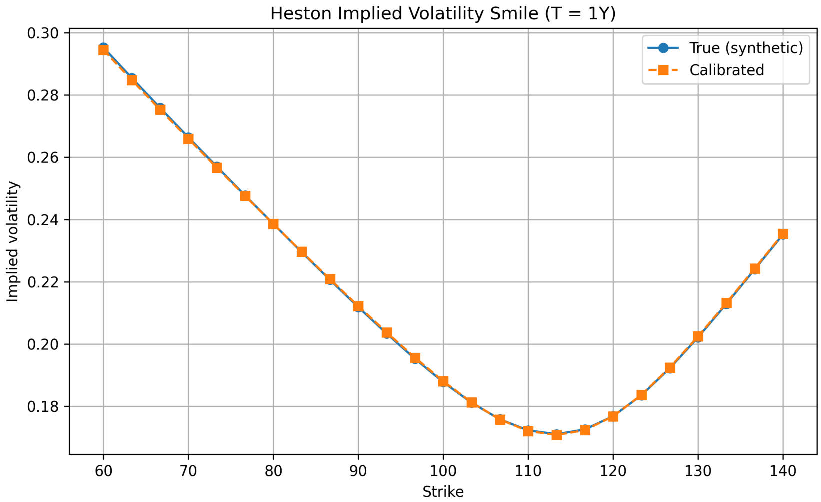

Figure 1: Implied volatility smile at maturity $T = 1$ year. The calibrated Heston model reproduces the synthetic smile almost exactly across strikes.

7.3 Surface-level agreement

Figure 2 shows a comparison between the full synthetic implied volatility surface and the surface implied by the calibrated Heston model. As expected in this noise-free setting, the two surfaces coincide up to numerical precision across strikes and maturities.

This highlights a well-known feature of Heston calibration: multiple parameter combinations may produce nearly identical implied volatility surfaces, leading to non-uniqueness of calibrated parameters even when surface-level fit is excellent.

7.4 Residual diagnostics

To assess the structure of remaining errors, Figure 3 reports a heatmap of implied volatility residuals, $\widehat{\sigma}_{\text{model}} - \sigma_{\text{synthetic}}$, across strikes and maturities. Residuals are at numerical precision (on the order of 10^{-8}) and exhibit no oscillatory, maturity-dependent, or moneyness-dependent structure.

This behaviour is expected in a noise-free synthetic experiment and confirms that the pricing, inversion, and optimisation components of the calibration pipeline are internally consistent. In practical calibration to market data, residual structure typically reflects both observation noise and model misspecification.

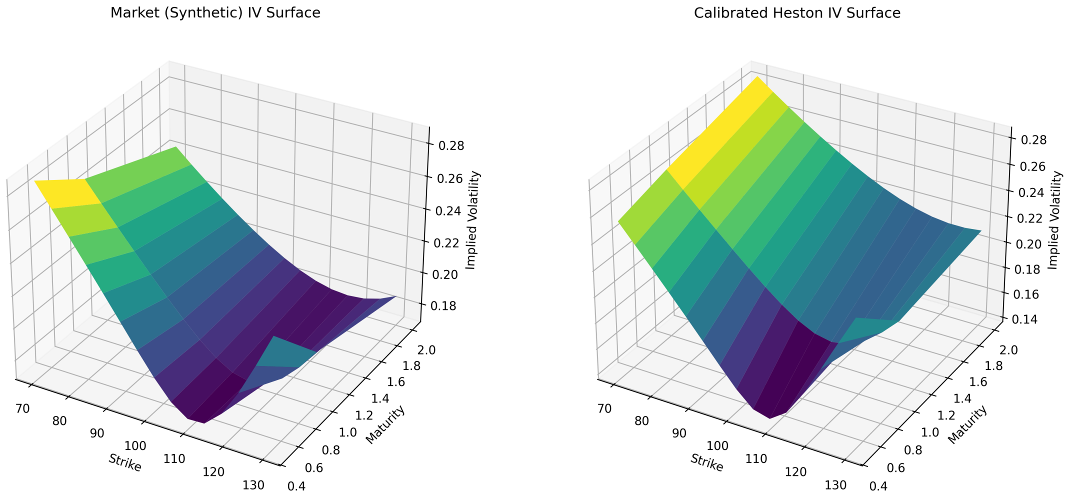

Figure 2: Synthetic (left) and calibrated Heston (right) implied volatility surfaces. The calibrated model closely reproduces the synthetic surface across strikes and maturities.

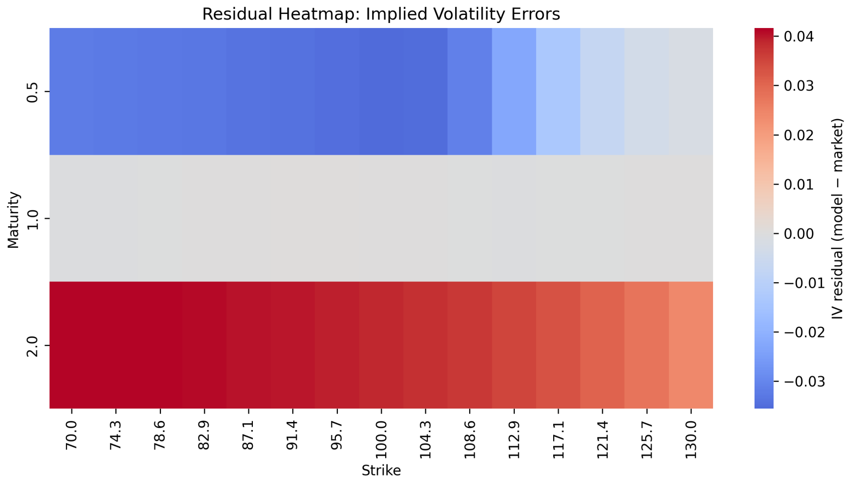

Figure 3: Residual heatmap of implied volatility errors (model minus synthetic market). Residuals are small and largely structureless across strikes and maturities.

7.5 Discussion

These experiments demonstrate that the proposed calibration framework recovers the data-generating implied volatility surface with numerical precision in a controlled, noise-free setting. This provides a strong verification of the correctness and internal consistency of the pricing, implied-volatility inversion, and optimisation components.

In practical calibration to market data, exact parameter recovery is neither expected nor desirable due to observation noise, bid–ask effects, limited strike coverage, and model misspecification. Under such conditions, multiple parameter configurations may produce similar surface-level fits, and diagnostics such as residual structure and parameter stability become essential.

While the Heston model provides a robust global fit, it is known to struggle with extremely steep short-dated skews, multi-modal smiles, or features induced by jumps. In practice, such limitations motivate the use of extensions including Heston–Hull–White hybrid models [7], stochastic-local volatility models [8], or rough volatility formulations [9].

8 Conclusion

This note has developed a complete and internally consistent framework for calibrating the Heston stochastic volatility model, spanning risk-neutral dynamics, characteristic function construction, Fourier-based pricing, and nonlinear least-squares optimisation. Particular emphasis has been placed on numerical stability, reproducibility, and the practical considerations required for reliable calibration.

Controlled numerical experiments using synthetic option data demonstrate that the proposed pipeline is able to reproduce implied volatility smiles and surfaces accurately across strikes and maturities. Diagnostic analysis highlights a well-known but important feature of Heston calibration: multiple parameter vectors may generate nearly identical implied volatility surfaces, leading to non-uniqueness of calibrated parameters despite excellent surface-level fit. Residual patterns remain small and stable, with deviations consistent with the structural limitations of the single-factor variance process.

The focus of this work is on the calibration methodology and numerical structure rather than on providing a line-by-line software implementation, which necessarily depends on language- and library-specific design choices. Nevertheless, the framework presented here is directly aligned with practitioner implementations and provides a transparent reference for quantitative research, model validation, and comparative studies.

Natural extensions include the incorporation of jumps, hybrid equity–interest rate dynamics, and more flexible volatility models, as well as the use of adjoint or algorithmic differentiation techniques to accelerate large-scale calibration and risk computations.

A Affine Characteristic Function Derivation

This appendix presents the derivation of the affine characteristic function

$$\phi(u;T) = \mathbb{E}^{\mathbb{Q}}\left[e^{iuX_T} \mid X_0, v_0\right], \qquad X_t = \ln S_t,$$

for the Heston stochastic volatility model.

For notational simplicity, we assume zero dividends in this appendix; extension to a constant dividend yield q is immediate via the standard drift adjustment.

A.1 State Dynamics

Under the risk-neutral measure $\mathbb{Q}$, the log-price and variance follow.

Applying Itô's lemma to the log-price process $X_t = \log S_t$ under the risk-neutral dynamics of S_t yields the following stochastic differential equation, where the term $-\frac{1}{2}v_t$ arises from the quadratic variation of S_t.

$$dX_t = \left(r - \frac{1}{2}v_t\right) dt + \sqrt{v_t}\, dW_t^{(1)}, \tag{9}$$

$$dv_t = \kappa(\bar{v} - v_t)\, dt + \sigma\sqrt{v_t}\, dW_t^{(2)}, \tag{10}$$

with instantaneous correlation $\mathrm{d}W_t^{(1)}\, \mathrm{d}W_t^{(2)} = \rho\, dt$.

Let the maturity be T and define $\tau = T - t$. We introduce the conditional characteristic function

$$f(t,x,v;u) := \mathbb{E}^{\mathbb{Q}}\left[e^{iuX_T} \mid X_t = x,\ v_t = v\right].$$

A.2 Affine Ansatz

We seek an exponentially affine solution of the form

$$f(t,x,v;u) = \exp(A(\tau,u) + B(\tau,u)v + iux), \qquad A(0,u) = 0,\ B(0,u) = 0. \tag{11}$$

A.3 Backward Kolmogorov Equation

The conditional characteristic function

$$f(t,x,v;u) = \mathbb{E}^{\mathbb{Q}}\big[e^{iuX_T} \mid X_t = x,\ v_t = v\big]$$

satisfies the backward Kolmogorov equation

$$-\frac{\partial f}{\partial \tau} + \mathcal{L}f = 0, \qquad f(0,x,v;u) = e^{iux}, \tag{12}$$

where $\tau = T - t$ and $\mathcal{L}$ denotes the infinitesimal generator of the two-dimensional diffusion (X_t, v_t) defined by (9)–(10).

The generator $\mathcal{L}$ acts on sufficiently smooth functions $f(x,v)$ as

$$\mathcal{L}f = \big(r - \tfrac{1}{2}v\big)\frac{\partial f}{\partial x} + \kappa(\bar{v} - v)\frac{\partial f}{\partial v} + \tfrac{1}{2}v\frac{\partial^2 f}{\partial x^2} + \rho\sigma v\frac{\partial^2 f}{\partial x\,\partial v} + \tfrac{1}{2}\sigma^2 v\frac{\partial^2 f}{\partial v^2}.$$

We substitute the exponentially affine ansatz

$$f(t,x,v;u) = \exp(A(\tau,u) + B(\tau,u)v + iux)$$

into the backward equation (12). The required derivatives are

$$\begin{aligned} f_x &= iuf, & f_{xx} &= -u^2 f,\\ f_v &= Bf, & f_{vv} &= B^2 f,\\ f_{xv} &= iuBf, & f_\tau &= (A_\tau + vB_\tau)f. \end{aligned}$$

Substituting these expressions into $-\partial_\tau f + \mathcal{L}f = 0$ and dividing through by f yields

$$0 = -A_\tau - vB_\tau + iu\big(r - \tfrac{1}{2}v\big) + \kappa(\bar{v} - v)B - \tfrac{1}{2}u^2 v + \rho\sigma iu\, vB + \tfrac{1}{2}\sigma^2 vB^2.$$

Separating terms independent of v gives

$$A_\tau(\tau,u) = iur + \kappa\bar{v}\,B(\tau,u),$$

while collecting coefficients of v yields the Riccati equation

$$B_\tau(\tau,u) = \tfrac{1}{2}\sigma^2 B^2 + (\rho\sigma iu - \kappa)B - \tfrac{1}{2}(u^2 + iu), \tag{13}$$

with initial condition $B(0,u) = 0$.

A.4 Solution of the Riccati Equation

The Riccati equation

$$B_\tau(\tau,u) = \gamma B^2 + \beta(u)B + \alpha(u), \qquad B(0,u) = 0,$$

with

$$\alpha(u) = -\tfrac{1}{2}(u^2 + iu), \qquad \beta(u) = \rho\sigma iu - \kappa, \qquad \gamma = \tfrac{1}{2}\sigma^2,$$

admits a closed-form solution.

Define the discriminant

$$d(u) = \sqrt{\beta(u)^2 - 4\alpha(u)\gamma} = \sqrt{(\rho\sigma iu - \kappa)^2 + \sigma^2(u^2 + iu)}.$$

Expanding the square shows

$$d(u)^2 = \kappa^2 + \sigma^2(1 - \rho^2)u^2 + iu\sigma(\sigma - 2\rho\kappa),$$

which is consistent with standard references.

We define

$$g(u) = \frac{\beta(u) - d(u)}{\beta(u) + d(u)}.$$

The solution for $B(\tau, u)$ is then given by

$$B(\tau, u) = \frac{\beta(u) - d(u)}{\sigma^2} \cdot \frac{1 - e^{-d(u)\tau}}{1 - g(u)e^{-d(u)\tau}}. \tag{14}$$

Integrating $A_\tau(\tau, u) = iur + \kappa\bar{v}B(\tau, u)$ yields

$$A(\tau, u) = iur\tau + \frac{\kappa\bar{v}}{\sigma^2}\left[(\beta(u) - d(u))\tau - 2\ln\left(\frac{1 - g(u)e^{-d(u)\tau}}{1 - g(u)}\right)\right]. \tag{15}$$

Branch selection and numerical stability

The complex square root $d(u)$ is evaluated using a branch chosen such that $\mathrm{Re}(d(u)) > 0$, commonly referred to as the *Gatheral convention*. This choice ensures continuity of the characteristic function across maturities.

However, when $1 - g(u) = 0$, the logarithmic term in (15) becomes singular. While this singularity is removable in exact arithmetic, it may lead to numerical instability in finite-precision implementations.

To mitigate this issue, alternative but algebraically equivalent formulations are often employed in practice. In particular, the *Little Trap* formulation of the Heston characteristic function avoids this singularity by rewriting the logarithmic term in a numerically stable form. Both representations yield identical prices when evaluated consistently, but the Little Trap formulation is generally preferred in calibration routines for its superior numerical robustness.

A.5 Final Characteristic Function

The characteristic function is therefore

$$\phi(u; T) = \exp\Big(A(T, u) + B(T, u)v_0 + iuX_0\Big).$$

For numerical implementation, the complex square root $d(u)$ is evaluated using a continuous branch chosen such that $\mathrm{Re}(d(u)) > 0$, ensuring stability and continuity across maturities. This convention is essential for reliable Fourier-based pricing and calibration.

B COS Method for European Option Pricing

This appendix summarises the Fourier–Cosine (COS) method of Fang and Oosterlee [1] for pricing European options under models with known characteristic functions, such as the Heston model. The COS method is included as an efficient and numerically stable alternative to FFT-based pricing, and is not essential to the calibration framework presented in the main text.

B.1 COS pricing framework

Let $X_T = \log S_T$ and denote its characteristic function by

$$\phi(u; T) = \mathbb{E}^{\mathbb{Q}}\left[e^{iuX_T}\right].$$

The price of a European call option with strike K and maturity T can be written as

$$C = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\left[(S_0 e^{X_T} - K)^+\right].$$

The COS method evaluates this expectation by expanding the payoff function in a cosine series over a truncated interval $[a, b]$ that captures the bulk of the probability mass of X_T. No explicit density reconstruction is required; all coefficients are computed directly from the characteristic function.

B.2 COS expansion

The COS method approximates European option prices by expanding the payoff function in a cosine series over a truncated log-price interval $[a, b]$. The resulting pricing formula for a European call option is

$$C_{\text{COS}} = e^{-rT} \sum_{k=0}^{N-1} \text{Re}\left[\phi\left(\frac{k\pi}{b-a}; T\right) e^{-ik\pi a/(b-a)}\right] V_k, \tag{16}$$

where N denotes the number of cosine terms, $\phi(u; T)$ is the characteristic function of $X_T = \log S_T$, and V_k are payoff-dependent cosine coefficients.

For a European call option with payoff $(S_T - K)^+$, the coefficients V_k are defined by

$$V_k = \frac{2}{b-a} \int_{\ln K}^{b} (e^x - K) \cos\left(\frac{k\pi(x-a)}{b-a}\right) dx.$$

These integrals admit closed-form expressions obtained by analytic evaluation, which depend explicitly on the truncation interval $[a, b]$ and the strike K. The resulting formulas are given in Fang and Oosterlee [1, Section 3] and are implemented directly in numerical COS pricing routines. For $k = 0$, the corresponding expressions are understood in the limiting sense to avoid division by zero.

An analogous construction applies to European put options. In the present work, the COS method is included as an alternative Fourier-based pricing approach; the explicit payoff coefficients are therefore not reproduced here, as they do not affect the calibration framework developed in the main text.

Cumulants for the Heston model

For the COS truncation interval, the first cumulants of the log-price $X_T = \log S_T$ under the Heston model are required. These may be obtained from derivatives of the log-characteristic function. For completeness, the first cumulant is

$$c_1 = \log S_0 + (r - q)T + \frac{1 - e^{-\kappa T}}{\kappa}(v_0 - \bar{v}),$$

while the second and fourth cumulants admit closed-form expressions involving κ, $\bar{v}$, σ, ρ, and T, and are reported in Fang and Oosterlee [1, Appendix A] and Grzelak and Oosterlee [7].

B.3 Choice of truncation interval

The truncation interval $[a, b]$ is chosen using cumulants of X_T:

$$a = c_1 - L\sqrt{c_2 + \sqrt{c_4}}, \qquad b = c_1 + L\sqrt{c_2 + \sqrt{c_4}},$$

where c_1, c_2, and c_4 denote the first, second, and fourth cumulants of X_T, respectively, and L is a user-selected parameter typically in the range $[8, 12]$. For the Heston model, these cumulants are available in closed form via derivatives of the log-characteristic function.

B.4 Remarks on calibration use

The COS method exhibits exponential convergence in N for smooth payoff functions and is highly efficient in calibration settings. In practice, COS and FFT-based Carr–Madan pricing yield comparable accuracy when numerical parameters are chosen consistently. The calibration framework developed in the main text is agnostic to the choice of pricing engine, and COS may be substituted directly for FFT-based pricing without modification to the optimisation procedure.

C Carr–Madan FFT Method for European Option Pricing

This appendix summarises the Carr–Madan fast Fourier transform (FFT) method for pricing European options [4]. The method is particularly well suited to calibration problems, as it enables simultaneous pricing across a large grid of strikes with high numerical efficiency.

C.1 Damped call price formulation

Let $k = \log K$ denote the log-strike and define the discounted European call price

$$C(k) = e^{-rT}\,\mathbb{E}^{\mathbb{Q}}\left[(S_T - e^k)^+\right].$$

To ensure square integrability of the pricing function, Carr and Madan introduce an exponential damping factor $\alpha > 0$ and define the damped call price

$$c_\alpha(k) = e^{\alpha k} C(k). \tag{17}$$

For sufficiently large α, the function $c_\alpha(k)$ belongs to $L^2(\mathbb{R})$, allowing Fourier transform techniques to be applied.

Throughout this appendix we work with spot prices; an equivalent forward-price formulation may be obtained by the usual change of numeraire.

C.2 Fourier transform representation

The Fourier transform of the damped call price is defined by

$$\hat{c}_\alpha(u) = \int_{-\infty}^{\infty} e^{iuk} c_\alpha(k)\,dk.$$

Carr and Madan show that this transform admits the closed-form expression

$$\hat{c}_\alpha(u) = \frac{e^{-rT}\,\phi(u - (\alpha+1)i; T)}{\alpha^2 + \alpha - u^2 + i(2\alpha+1)u}, \tag{18}$$

where $\phi(u;T)$ denotes the characteristic function of $\log S_T$. The Carr–Madan formulation requires evaluation of the characteristic function at complex arguments of the form $u - (\alpha+1)i$. Consequently, the Riccati solution underlying $\phi(u;T)$ must be analytically continued to complex frequencies. In this setting, the choice of branch for the complex square root in the discriminant $d(u)$ becomes particularly important, as inconsistent branch selection across the FFT grid can lead to discontinuities and numerical instability.

In practice, the branch is chosen such that $\mathrm{Re}(d(u)) > 0$ for all relevant complex arguments, and numerically stable formulations (such as the Little Trap representation) are often employed to ensure continuity of the characteristic function when pricing via FFT.

C.3 Inverse transform and FFT discretisation

The damped call price can be recovered via the inverse Fourier transform

$$c_\alpha(k) = \frac{1}{2\pi} \int_{-\infty}^{\infty} e^{-iuk}\, \hat{c}_\alpha(u)\, du. \tag{19}$$

To evaluate (19) numerically, the integration variable is discretised on a uniform grid

$$u_j = j\eta, \qquad j = 0, \dots, N-1,$$

with corresponding log-strike grid

$$k_m = k_{\min} + m\,\Delta k, \qquad m = 0, \dots, N-1,$$

where the spacing satisfies

$$\Delta k = \frac{2\pi}{N\eta}.$$

With this choice, the inverse transform may be evaluated efficiently using the FFT, yielding all values $\{c_\alpha(k_m)\}_{m=0}^{N-1}$ in $O(N \log N)$ time. Numerical quadrature weights (e.g. Simpson's rule) are commonly applied to improve accuracy.

C.4 Recovery of option prices

Once the damped prices are computed, undiscounted call prices are recovered via

$$C(K_m) = e^{-\alpha k_m}\, c_\alpha(k_m), \qquad K_m = e^{k_m}.$$

This produces a vector of option prices across a wide strike range from a single FFT evaluation.

C.5 Choice of numerical parameters

Stable FFT-based pricing requires careful but fixed selection of numerical parameters:

- **Damping factor:** $\alpha \in [1, 3]$ is typical; it must be held fixed throughout calibration.
- **Frequency spacing:** η determines the integration range $u_{\max} = N\eta$.
- **Grid size:** N is chosen as a power of two (e.g. 2^{12} or 2^{13}).
- **Strike range:** controlled via $k_{\min}$ and Δk.

All numerical parameters are fixed across optimisation iterations to avoid introducing artificial noise into the calibration objective.

C.6 Remarks on calibration use

The Carr–Madan FFT method is extremely efficient for large strike grids and is therefore well suited to nonlinear calibration routines requiring repeated pricing evaluations. When numerical parameters are chosen consistently, FFT-based pricing yields smooth objective functions compatible with both gradient-based and derivative-free optimisation algorithms.

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